Jiayao Li

Email: 2351405@tongji.edu.cn | Tel: +86 132 9198 9727 | HomePage | GitHub

Education

Tongji University **Shanghai, CN**
***Bachelor** of Computer Science and Technology (Elite Class)* GPA: 4.63/5 Rank: 27/125 CET-6: 582 09/2023 - 06/2027
Research interests: Embodied AI & Robotics; Physical-World AI & Simulation; 3D Perception & Spatial Understanding

Selected Honors

Qidi Scholarship	2024-2025	School-wide, top 1.2%
Guo Xie Birong Scholarship	2024-2025	School-wide, First-Class, top 1.7%
Social Activity Scholarship & Outstanding Student Cadre	2023-2024 / 2024-2025	School-wide

Publications

Two-Stream Interactive Joint Learning of Scene Parsing and Geometric Vision Tasks Feb. 2026
*Guanfeng Tang, Hongbo Zhao, Ziwei Long, **Jiayao Li,** Bohong Xiao, Wei Ye, Hanli Wang, Rui Fan*
Under review at **IEEE Transactions on Intelligent Vehicles. arXiv preprint.** [arXiv] [PDF]

Are LLMs Reliable for Recommending Libraries and Their Usage? An Empirical Study May 2026
*Yuying Wang, Yilian Yu, **Jiayao Li,** Maojun Huang, et al.*
Under review at Internetware 2026 (CCF C).

Can video generation replace cinematographers? Research on the cinematic language of generated video Dec. 2024
*Xiaozhe Li, Kai Wu, Siyi Yang, YiZhan Qu, Guohua Zhang, Zhiyu Chen, **Jiayao Li** et al.* **arXiv preprint.** [arXiv] [PDF]

Research Experiences

Scene Parsing and Geometric Vision — Two-Stream Interactive Joint Learning Framework Sep.2025 – Dec.2025
Advisor: *Prof. Rui Fan, Tongji University*

- **Project Description**: Developed **a two-stream interactive joint learning framework for scene parsing and geometric vision tasks**. The framework enhances segmentation performance by incorporating cross-view geometric information from the geometric vision stream, improving semantic segmentation accuracy on *vKITTI2, Cityscapes*, and *KITTI 2015*.
- **Personal Contributions**: Conducted **benchmark evaluation** and visualization analysis across multiple models, including *DFormerv2*. **Experimentally validated the effectiveness** of *the Cross-Task Adapter* for feature fusion. *TwInS* achieved up to a 7.23% improvement in semantic segmentation *mIoU* and up to a 13.64% improvement in instance segmentation *mAP*.
- **Outcome**: Fourth author of the manuscript *Two-Stream Interactive Joint Learning of Scene Parsing and Geometric Vision Tasks*, currently under review at **IEEE Transactions on Intelligent Vehicles**.

Single-View 3D Occupancy Prediction and Spatial Perception — ViPOcc Framework Sep.2025 – Dec.2025
Advisor: *Prof. Rui Fan, Tongji University*

- **Project Description**: Developed **a single-view 3D occupancy prediction framework**, leveraging depth and semantic-instance priors from *visual foundation models*. Designed *inverse-depth alignment* and *Semantic-guided Non-overlapping Gaussian Mixture Sampling (SNOG)* to improve 3D spatial perception from monocular RGB images.
- **Personal Contributions**: Conducted **functional validation** of *the inverse-depth alignment module* and *the SNOG sampler*. Quantitatively improved performance, including a 2.9% increase in invisible-region accuracy, a 6.7% increase in object-level invisible-region recall, and a 6.2% reduction in Abs Rel for depth estimation. Led the **organization of the technical solution for the national competition** and optimized the presentation of the project methodology.
- **Outcome**: Second Prize, 7th Global Artificial Intelligence Algorithm Elite Competition, 2025.

Dynamic Robot Navigation — Optimization of 2D LiDAR-Based Pedestrian Detection Sep.2025 – Jan.2026
Advisor: *Mohan Jiang, Shanghai Innovation Institute*

- **Project Description**: Developed **a dynamic navigation system** integrating perception and decision-making for mobile robot navigation in dynamic environments. The perception module was optimized based on the *DR-SPAAM*, while the decision-making module adopted *TD3 reinforcement learning* with a *four-stage curriculum learning strategy*.
- **Personal Contributions**: **Optimized the perception module** by replacing *Mask R-CNN* with *YOLOv8* to generate higher-quality visual pseudo-labels for training the *DR-SPAAM* detector. Improved the pedestrian detection metric *AP@0.5m* by 35.7%, from 0.42 to 0.57. Developed a systematic understanding of the **perception–decision–control closed loop**.

Vision-and-Language Navigation, VLN — Literature Review and Exploratory Study Mar.2026 – Present

- Conducted a systematic review of recent VLN studies, including *DecoVLN, AgentVLN*, and reproduced selected open-source codebases. Summarized key challenges in long-horizon navigation, including **cross-modal instruction grounding, memory compression, streaming decision-making**, and **closed-loop error correction** , with a focus on VLM-as-Brain paradigms.

Selected Competitions

Outstanding Winner & INFORMS Award (Top Prize, 0.12%), Mathematical Contest in Modeling	May 2026
Second Prize, National Artificial Intelligence Application Scenario Innovation Challenge	Jan. 2026
Second Prize, Final Round of the 7th Global Campus Artificial Intelligence Algorithm Elite Competition	Dec. 2025
Third Prize, Lanqiao Cup National Software and Information Technology Professionals Competition	May 2025

Technical Skills

- **Tech Stack (selected):** C/C++, Python, Java; PyTorch / TensorFlow; Django+Vue 3 (Full-Stack); OpenCV; Verilog; Git.
- **Social Activities (selected):** Volunteer for the 8th China International Import Expo (CIIE); Class Monitor.